

How a lung cell processes and displays an MHC-I peptide

LUNG | 7 TISSUE-HLA TASKS

Representative Human context • six biological layers • canonical genes shown for orientation

BIOLOGICAL LAYERS

1

Source abundance & lung cell state

EPCAM SFTPA1 SFTPB SCGB1A1

2

Proteasomal cleavage

PSMB8 PSMB9 PSMB10

3

Transport into the ER

TAP1 TAP2

4

ER peptide trimming

ERAP1 ERAP2

5

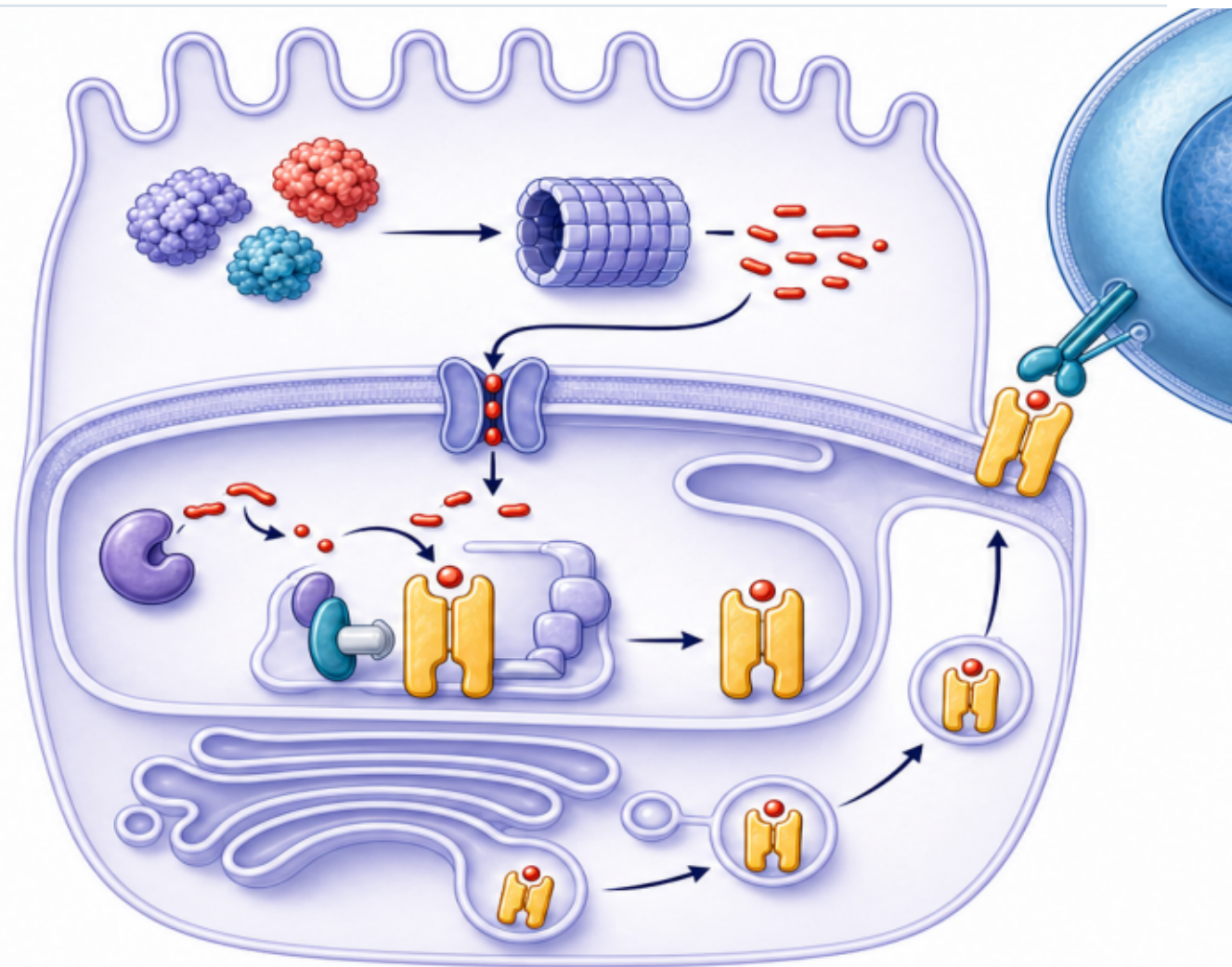
Peptide-loading complex

TAPBP CALR PDIA3

6

Surface MHC-I display

HLA-A/B/C B2M



SOURCE → CLEAVAGE → ER TRANSPORT → TRIMMING → LOADING → SURFACE DISPLAY